Discrete Mathematics

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FUNDAMENTALS OF LOGIC

MATHEMATICAL INDUCTION

Let $P(n)$ be a statement for all positive integers.

$P(n)$ is true for all positive integers, if :

1. $P(1)$ is true
2. For all $k \geq 1$, $P(k) \rightarrow P(k+1)$.

If $P(1)$ is true, then:
$P(1) \Rightarrow P(2) \Rightarrow P(3) \Rightarrow \ldots$

Some statements may fail for initial values but hold for $n \geq n_0$.

To prove $P(n)$ for all $n \geq n_0$, start the base case at $P(n_0)$.

MATHEMATICAL INDUCTION

1. $P(n_0)$ is true.
2. For all $k \geq n_0$, $P(k) \rightarrow P(k+1)$.

Steps involved in a proof based on mathematical induction

Step 1: (Base of induction): Show that $P(n_0)$ is true.

Step 2 (Induction Hypothesis): Assume $P(k)$ is true, i.e. the statement as got by putting '$n = k$' is true.

Step 3 (Inductive step): Show $P(k+1)$ is true

MATHEMATICAL INDUCTION

Example: 1
Conjecture
Example: 2
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Example: 4
Strong mathematical induction
Recursion
Example: 5
Example: 6

Example: 1

Show that if 'n' is a positive integer, then

$$1 + 2 + 3 + \ldots + n = \frac{n(n+1)}{2}.$$

MATHEMATICAL INDUCTION

Example: 1

Show that if 'n' is a positive integer, then

$$1 + 2 + 3 + \ldots + n = \frac{n(n+1)}{2}.$$

Let $P(n)$: sum of first n positive integers $= \frac{n(n+1)}{2}$.

- To prove by induction:
- Base case: $P(1)$ is true.
- Inductive step: Show $P(k) \Rightarrow P(k+1)$ for $k = 1, 2, 3, \ldots$

Step1: Basis of induction

Now $P(1)$ is true because $1 = \frac{1(1+1)}{2}$; $P(n) = \frac{n(n+1)}{2}$ is true for $n = 1$.

Step2: Induction Hypothesis

Assume $P(k) = \frac{k(k+1)}{2}$ is true for $k \geq 1$.

MATHEMATICAL INDUCTION

Show $P(k+1)$ is true.

Step3: Inductive step

$$p(k+1) = 1+2+3+\ldots+k+k+1 = (1+2+3+\ldots+k)+k+1$$
$$= \frac{k(k+1)}{2} + (k+1) = \frac{k(k+1)+2(k+1)}{2}$$
$$= \frac{(k+1)(k+2)}{2} = \frac{(k+1)[(k+1)+1]}{2}$$

So, based on the induction hypothesis in step2, this means

$$P(k) = \frac{k(k+1)}{2} \rightarrow P(k+1) = \frac{(k+1)((k+1)+1)}{2}$$

Hence, the proposition $P(n)$ is true.

MATHEMATICAL INDUCTION

Conjecture: An opinion or conclusion formed on the basis of incomplete information.

Example: 2

Conjecture a formula for the sum of the first n positive odd integers.

— The sum of first n positive odd integers for n = 1, 2, 3, 4, 5 are

$$
\begin{aligned}
1 &= 1 \\
1+3 &= 4 \\
1+3+5 &= 9 \\
1+3+5+7 &= 16 \\
1+3+5+7+9 &= 25
\end{aligned}
$$

MATHEMATICAL INDUCTION

the sum of first 'n' positive odd integers is n^2 i.e.

$$1 + 3 + 5 + \ldots + (2n - 1) = n^2$$

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$$1 + 3 + 5 + \ldots + (2n - 1) = n^2$$

Let $P(n)$ denote the proposition that the sum of first n positive odd integers is n^2.

Step 1: Basis of induction

$$P(1) : 1 = (1)^2$$
So, $P(1)$ is true
$P(n) = n^2$ is true for $n = 1$

Step 2: Inductive Hypothesis

For $k \geq 1$, $P(k) = k^2$ is true.

MATHEMATICAL INDUCTION

Step 3: Inductive step

$$
\begin{aligned}
P(k+1) &= 1+3+5+\ldots+(2k-1)+(2k+1) \\
&= \{1+3+5+\ldots+(2k-1)\}+(2k+1) \\
&= k^2+(2k+1) \\
&= (k+1)^2
\end{aligned}
$$

So, based on the induction hypothesis in step 2, it means

$$P(k) = k^2 \longrightarrow P(k+1) = (k+1)^2$$

Hence the proposition $P(n)$ is true.

MATHEMATICAL INDUCTION

Example: 3

Find and prove a formula for the sum of the first n cubes i.e., $1^3 + 2^3 + 3^3 + \ldots + n^3$.

MATHEMATICAL INDUCTION

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Find and prove a formula for the sum of the first n cubes i.e., $1^3 + 2^3 + 3^3 + \ldots + n^3$.

Now,

$$\begin{aligned} 1^3 &= 1 = 1^2 \\ 1^3 + 2^3 &= 9 = 3^2 \\ 1^3 + 2^3 + 3^3 &= 36 = 6^2 \\ 1^3 + 2^3 + 3^3 + 4^3 &= 100 = 10^2 \end{aligned} \quad \Bigg| \quad \begin{aligned} 1 &= 1 \\ 1 + 2 &= 3 \\ 1 + 2 + 3 &= 6 \\ 1 + 2 + 3 + 4 &= 10 \end{aligned}$$

Conjecture: $1^3 + 2^3 + \ldots + n^3 = \left(\frac{n(n+1)}{2}\right)^2$

MATHEMATICAL INDUCTION

Example: 4

Prove that if $n \geq 2$, the generalized De Morgan's law

$$\sim (p_1 \wedge p_2 \wedge \cdots \wedge p_n) \equiv (\sim p_1) \vee (\sim p_2) \vee \cdots \vee (\sim p_n) \text{ is true.}$$

MATHEMATICAL INDUCTION

Example: 4

Prove that if $n \geq 2$, the generalized De Morgan's law

$$\sim (p_1 \wedge p_2 \wedge \cdots \wedge p_n) \equiv (\sim p_1) \vee (\sim p_2) \vee \cdots \vee (\sim p_n) \text{ is true.}$$

For $n = 1$, is meaningless (as De Morgan's laws deal with 2 or more variables)

For $n = 2$:

$$\sim (p_1 \wedge p_2) \equiv (\sim p_1) \vee (\sim p_2) \text{ is true.}$$

$P(k) : \sim (p_1 \wedge p_2 \wedge \cdots \wedge p_k) \equiv (\sim p_1) \vee (\sim p_2) \vee \cdots \vee (\sim p_k)$

Step1: Basis of induction

$P(2)$ is true (De Morgan's Law).

Step2: Induction Hypothesis

$P(k)$ is true, i.e.

$$\sim (p_1 \wedge p_2 \wedge \cdots \wedge p_k) \equiv (\sim p_1) \vee (\sim p_2) \vee \cdots \vee (\sim p_k)$$

MATHEMATICAL INDUCTION

Step3: Inductive step

$$\begin{aligned} P(k+1): &\sim (p_1 \wedge p_2 \wedge \cdots \wedge p_k \wedge p_{k+1}) \\ \equiv &\sim \{(p_1 \wedge p_2 \wedge \cdots \wedge p_k) \wedge p_{k+1}\} \end{aligned}$$

We can put a bracket enclosing $p_1 \ldots p_k$, because we know that '$\wedge$' is an associative.

$$\equiv \sim (p_1 \wedge p_2 \wedge \cdots \wedge p_k) \vee (\sim p_{k+1})$$

$$= \sim p_1 \vee \sim p_2 \vee \cdots \vee \sim p_k \vee \sim p_{k+1}$$

So, $P_k \rightarrow P_{k+1}$ for $k \geq 2$
Hence $P(n)$ is true, $\forall n \geq 2$

MATHEMATICAL INDUCTION

Strong mathematical induction

Let $P(n)$ be a statement for integers. $P(n)$ is true for all positive integers if $\exists q \geq 1$ such that :

1. $P(1), P(2), \ldots, P(q)$ are all true.
2. When $k \geq q$, the assumption that $P(i)$ is true $\forall\ \mathbb{Z}\ 1 \leq i \leq k$ implies that $P(k+1)$ is true.

The steps of proof are:

1. **Basis of induction:** Show that $P(1), P(2), \ldots, P(q)$ are all true.
2. **Strong inductive Hypothesis:** Assume that $P(i)$ is true for all integers $1 \leq i \leq K$, where $K \geq q$.
3. **Inductive step:** Show that $P(K+1)$ is true.

MATHEMATICAL INDUCTION

Recursion

Let 'n' be the set of non-negative integers.

A function on non-negative integers is defined recursively, if:

- $f(0)$ is given, and
- for each positive integer 'n' the value of $f(n)$ is defined in terms of the value of 'f' [when $0 \leq k \leq n$].

Example: 5

A typical example of recursion is the Fibonacci sequence, where

$$\left.\begin{array}{l} f_0 = 1 = f_1 \\ f_{n+1} = f_n + f_{n-1} \quad \forall\ n \geq 1 \end{array}\right\} \text{Fibonacci Sequence Recursively}$$

MATHEMATICAL INDUCTION

Let us start calculating the elements of the sequence.

$$\begin{array}{rcccccc} n & = & 0 & 1 & 2 & 3 & 4 \\ f(n) & = & f(0) & f(1) & f(0)+f(1) & f(2)+f(1) & f(3)+f(2) \\ f(n) & = & 1 & 1 & 1+1=2 & 2+1=3 & 3+2=5 \end{array}$$

It is known, the nth Fibonacci no. is given by

$$f(n) = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^{n+1} - \left(\frac{1-\sqrt{5}}{2}\right)^{n+1}\right], \ \forall n \geq 0$$

MATHEMATICAL INDUCTION

Proof:

$$n = 0, \qquad f_0 = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^1 - \left(\frac{1-\sqrt{5}}{2}\right)^1\right] = 1$$

$$n = 1, \qquad f_1 = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^2 - \left(\frac{1-\sqrt{5}}{2}\right)^2\right] = 1$$

2. Strong induction hypothesis:

For $k \geq 1$, assume that $\{k \geq n$ here is $n \geq 1$, we have $f(0), f(1)$ to be true.$\}$

$$f_k = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^{k+1} - \left(\frac{1-\sqrt{5}}{2}\right)^{k+1}\right]$$

for each integer k, where $0 \leq k \leq n$.

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Example: 6

Suppose that we have stamps of 2 different denominations of ₹3, ₹5. Show that it is possible to make up postage ₹8 or more by using the stamps of these denominations only.

MATHEMATICAL INDUCTION

Example: 6

Suppose that we have stamps of 2 different denominations of ₹3, ₹5. Show that it is possible to make up postage ₹8 or more by using the stamps of these denominations only.

$$
\begin{aligned}
8 &= 3 + 5 \\
9 &= 3 + 3 + 3 \\
10 &= 5 + 5 \\
11 &= 5 + 3 + 3 \\
12 &= 3 + 3 + 3 + 3
\end{aligned}
$$

So, using denominations of ₹3 & ₹5. prove this using mathematical induction.

Basis of induction: ₹8 postage can be made up of the denomination ₹3 & ₹5.

MATHEMATICAL INDUCTION

Induction Hypothesis: Postage of ₹k can be formed using ₹3 & ₹5, for $k \geq 8$.

Inductive step:

Case 1: Suppose atleast one ₹5 stamp is used for ₹k.$\rightarrow$ Replace one ₹5 with two ₹3 stamps $\Rightarrow$ postage $=$ ₹k $+ 1$.

Case 2: Again, if No ₹5 stamp used for ₹k ($k \geq 8$).
$\rightarrow$ Replace atleast three ₹3 stamps with two ₹5 stamps $\Rightarrow$ postage $=$ ₹k$+1$.

Therefore, $P(n)$ is true for all $n \geq 8$.

1. **Basis of induction:**
   $P(8), P(9), P(10), P(11), P(12)$ are true.

   Postages ₹8–₹12 can be formed using₹3 & ₹5 stamps.

MATHEMATICAL INDUCTION

2. **Strong Inductive Hypothesis:**
   Assume $P(i)$ true for $8 \leq i \leq k$, where $k \geq 12$.
3. **Inductive Step:** $P(k-4)$ is true $\Rightarrow$ postage $(k-4)$ can be formed.
   Add one ₹5 stamp $\Rightarrow$ postage ₹$(k+1)$.

$$k - 4 \geq 8$$
$$k - 4 + 5 \geq 8 + 5 \implies k + 1 \geq 13$$

Since $k - 4 \geq 8 \Rightarrow k + 1 \geq 13$, $P(k+1)$ is true.

Hence, $P(n)$ is true for all $n \geq 8$.

Thanks for your attention!